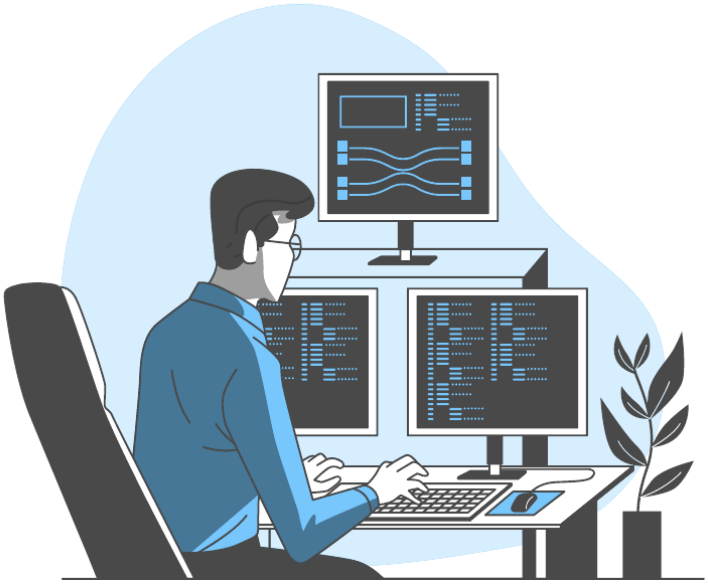
Local SRv6 Path Steering and Measured Failure Recovery

Project and Demo Report

Can SRv6 recover faster after a link failure?



TEAM SAGE

A3-PS020-TC231

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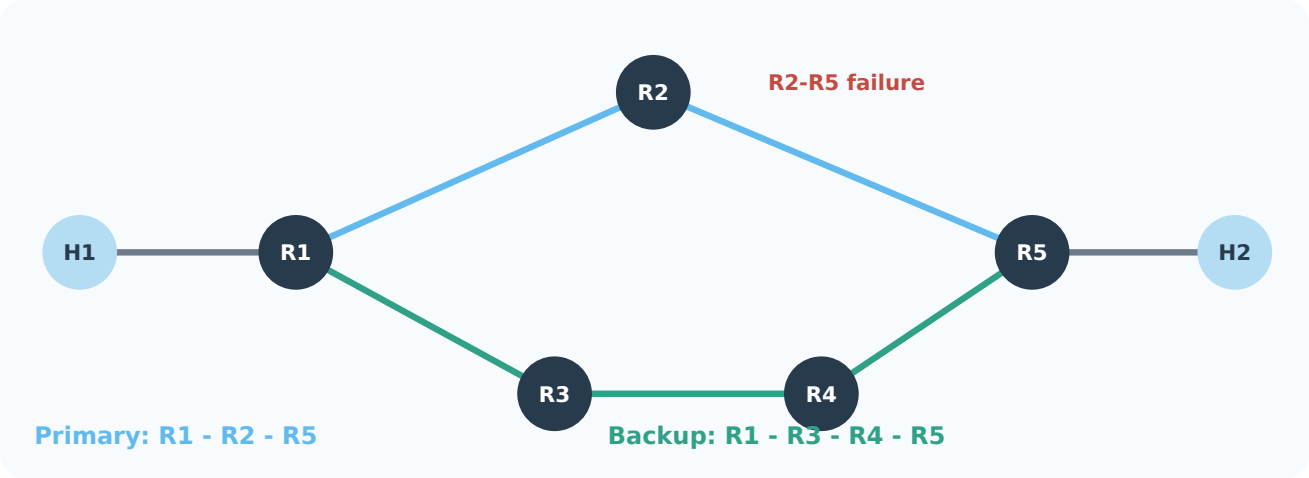
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50 trials per forwarding method

1. Problem and topology

Comparing recovery after a controlled R2-R5 link failure

We built a five-router IPv6 testbed to compare ordinary OSPFv3 forwarding with an SRv6 policy that steers traffic through a chosen sequence of routers.



- Baseline: IPv6 packets follow routes learned through OSPFv3.
- SRv6: R1 applies an explicit SID list and switches to a predefined backup policy.
- Test: H1 sends UDP traffic to H2 while R2 eth2 is brought down and later restored.

2. How failover works

SRv6 policy switching and restoration

- 1. Send H1 sends 1,000 UDP packets per second to H2. R1 starts on the primary SRv6 policy.
- 2. Detect After R2-R5 is disabled, R1 detects missing ICMPv6 probe replies using a 5 ms threshold and a 1 ms confirmation window.
- 3. Switch A script on R1 installs the backup policy via R3 and R4. R5 decapsulates the packets and forwards them to H2.
- 4. Restore The harness brings the link back, reinstalls the primary policy, checks connectivity and verifies fresh UDP delivery.

SID lists in forwarding order

Policy	Path	SID list
A	**R1 - R2 - R5**	**fc00:0:2::1, fc00:0:5::d6**
B	**R1 - R3 - R4 - R5**	**fc00:0:3::1, fc00:0:4::1,**
B continued		**fc00:0:5::d6**

The SRv6 method uses scripted ingress switching. It does not implement BFD or TI-LFA.

3. Final results

50 attempts per method, same topology and traffic profile

All 100 trials passed the functional checks. The table shows the mean for trials that also met the measurement-quality checks.

Measure	IPv6 / OSPFv3	SRv6
Functional passes	**50 / 50**	**50 / 50**
Accepted measurements	**47 / 50**	**46 / 50**
Packets lost at cut	**12.49**	**7.26**
Maximum receiver gap	**13.71 ms**	**8.40 ms**
UDP recovery from cut request	**13.42 ms**	**8.04 ms**

FEWER PACKETS LOST
42%

SHORTER RECEIVER GAP
39%

SHORTER UDP RECOVERY
40%

In this testbed, the SRv6 scripted switch recovered faster on average. The percentages compare the accepted-trial means against the IPv6 baseline.

The cut request starts the timing. Recovery is the first UDP packet received at H2 after the loss burst. The receiver gap includes normal packet spacing.

4. Prototype to final study

More trials and clearer measurement

Measure	Prototype	Final
Trials per method	**5 reported**	**50 attempted**
SRv6 packets lost	**5.8**	**7.26**
SRv6 UDP recovery	**7.72 ms**	**8.04 ms**
IPv6 UDP recovery	**13.81 ms**	**13.42 ms**

The final SRv6 figures are slightly higher than the prototype. The final comparison covers a larger sample and adds stricter checks, so we do not treat the two studies as identical.

What we improved

- Timing: Monotonic timestamps and a link-down request/ACK bracket make the measurement clearer.
- Detection: Reply validation, arming and confirmation checks reduce the chance of switching on an isolated missed reply.
- Validation: We record sender pacing, verify restoration and retain all attempted trials.

The prototype had two early triggers; their precise cause remains unresolved. The final campaign reported no pre-cut trigger in its 50 SRv6 trials.

5. Path verification

A separate captured failure and restoration trial

A captured SRv6 trial checked the Segment Routing Header as packets crossed the primary and backup paths, and again after restoration.

Observation	Result	Segments Left
Primary ingress at R1	**R2 SID, then R5**	**1**
After R2 at R5	**R5 SID active**	**0**
Backup ingress at R1	**R3, R4, then R5**	**2**
After R3 / after R4	**Next SID advances**	**1 / 0**
Restored path	**Primary path observed**	**1 then 0**

What the capture showed

- Backup delivery: The recovery packet followed policy B through R3 and R4 and reached the H2 link.
- Restoration: 100 fresh packets followed the restored primary path and reached H2.
- Header overhead: The capture showed 80 bytes for the two-SID primary and 96 bytes for the three-SID backup, including the outer IPv6 header.

These SID observations come from one captured trial. The two 50-trial campaigns ran without packet capture.

6. Conclusion

Results and next steps

In our five-router lab, the scripted SRv6 backup recovered UDP traffic in 8.04 ms on average, compared with 13.42 ms for the IPv6/OSPFv3 baseline.

What to keep in mind

- Measurement quality: 46 of 50 SRv6 and 47 of 50 IPv6 measurements met the quality checks. The remaining trials are retained in the results.
- Scope: This comparison uses one topology, one failure and one traffic rate on a WSL2 software testbed. It is not a general SRv6 performance guarantee.
- Recovery time: Probe recovery, UDP recovery and the policy-command duration describe different events. OSPF internal convergence was not isolated.

Future work

- Detector and pacing: Investigate early triggers and sender timing gaps, and test additional detector thresholds.
- Broader comparison: Repeat on native Linux and compare with BFD-assisted routing and TI-LFA.

The repository contains the source code, setup instructions, batch results and packet captures for the demonstration.